

Running eSim on Aakash tablets

Aakash tablets were introduced into the Indian market as an affordable platform for learning and education. Porting eSim into Aakash gives access to powerful CAD systems at a cost less than US\$ 40, thereby inspiring budding engineering minds who cannot afford costly hardware and software with brilliant ideas.

Software at a Glance

Name: eSim (previously known as Oscan/FreeEDA)

Type: EDA tool

Developers: Actively developed by FOSSEE group at IIT Bombay

Licence type: GNU general public licence

Key features:

- Draw circuits using KiCad, create a netlist and simulate using Ngspice
- KiCad for PCB design and Gerber files generation
- Model builder, Subcircuit Builder and SMCsim tools

Supporting platforms: Ubuntu Linux and Windows

Website: www.esim.fossee.in

Scilab based Mini Circuit Simulator (SMCsim). Electronic simulations are based on mathematical equations. In order to impart a clear idea of how the entire simulation actually works, it is necessary that mathematical equations are provided along with simulations. This unique feature of eSim helps users get a clear picture of simulations.

The tool runs in three modes. It solves circuit equations and gives the final simulation result in normal mode. Users can also view symbolic equations along with the result in symbolic mode. Numerical mode gives intermediate numerical values of elements and components in system matrices along with symbolic equations and the final result.

The other modules

We have already described the key features of the software. Let us now have a quick look at other important modules used in eSim software.

EEschema. This is an integrated software library where functionalities of circuit drawing, control, layout, library management and access to PCB design software are carried out. This schematic-capture software also provides the netlist describing the electrical connections of the circuit.

eSim developers have extended its functionality by incorporating software libraries for voltage and current sources.

CvPcb. This is basically a footprint editor tool that helps users to associate schematic components to component footprints in the PCB design.

Pcbnew. Pcbnew is the layout editor tool from KiCad. CvPcb tool assigns components for the netlist produced by EEschema to the module used by Pcbnew.

KiCad to Ngspice netlist converter. This module is developed by eSim developers to facilitate easy conversion of KiCad-generated netlists to Ngspice-compatible formats.

Analysis Inserter. This module helps in inserting the type and options for various kinds of analyses of electronic circuits. Various types of analysis such as operating point analysis, AC and DC analysis and transient analysis can all be done with the help of this tool developed for eSim.

Ngspice. This is yet another general-purpose circuit-simulation program for non-linear DC, non-linear transient and linear AC analyses. Simulation can contain components including resistors, capacitors, inductors, transmission lines BJTs, FETs,

metal oxide semiconductor field effect transistors, etc.

eSim workflow

The circuit diagram drawn on paper is entered into a computer using EEschema, which is the schematic editor used in this software. Apart from the library that provides voltage and current sources, there is yet another one created for printing and plotting purposes. The netlist file provided by this schematic editor contains electrical connections involved in the design.

Physical components are mapped into their footprints for creation of the PCB layout. CvPcb tool is used for this footprint mapping. The PCB layout is drawn with the help of Pcbnew. It is not always practical to check the accuracy of the design using breadboard testing. Model Builder helps in creating accurate models and editing existing models. Hierarchical modelling helps in designing complex models. Subcircuit Builder helps in creating sub-circuits.

Netlist converter helps in the conversion of netlists into a Ngspice-compatible format. Analysis Inserter proves to be an effective graphical interface for various types of simulations performed.

For analogue-, digital- and mixed-signal circuit simulation, eSim uses Ngspice software. SMCsim is also provided.

An indigenous open source software

There is a welcome trend in the country to promote the use of open source software, wherever possible. eSim was developed by the research group at IIT Bombay as a part of this initiative. The widespread use of open source software will certainly encourage a lot of people into serious research, especially in an economically-developing country like India where mastermind ideas are limited by shortage of funds. ●

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